

## Validity and reliability of the Indonesian version of QOLIE-10



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### ABSTRACT

**Background:** Epilepsy could result in disability and lowered patient's quality of life. Quality of Life in Epilepsy Inventory 10 (QOLIE-10) is a questionnaire to assess epilepsy patients' quality of life, but it has not been translated into Indonesian yet. This study aims to translate QOLIE-10 into Indonesian and analyze its validity and reliability.

**Methods:** This was a cross sectional study conducted at Neurology Clinic, Universitas Sebelas Maret, Surakarta. The translation process consists of two steps, a forward and backward process; each is done by two independent translators. Participants recruited were then asked to complete the Indonesian version of QOLIE-10 twice, with an interval of approximately one month.

**Results:** From the data obtained, we observed an r-value ranging from 0.247 to 0.669 on the pretest and post-test, greater than the critical r-value of 0.235 ( $p < 0.05$ ). We also observed Cronbach's alpha coefficient of 0.938 and 0.909 ( $p < 0.05$ ) for the pretest and post-test, respectively, and the test-retest reliability with  $r = 0.871$  ( $p = 0.000$ ).

**Conclusion:** Indonesian version of QOLIE-10 is understandable to Indonesian native speakers, valid, and reliable.

**Keywords:** Epilepsy, Indonesian, QOLIE, questionnaire, quality of life, reliability, validity.

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## INTRODUCTION

Epilepsy is a chronic neurological disease characterized by seizures that could affect people of all ages, races, social classes, and geographical locations.<sup>1</sup> International League Against Epilepsy (ILAE) defined epilepsy as at least two unprovoked seizures occurring more than 24 hours apart or one unprovoked seizure with a 60% probability of further seizures occurring over the next ten years or diagnosis of an epilepsy syndrome.<sup>2</sup> One study estimated that over 45 million people live with epilepsy worldwide, with a prevalence of 2.3-15.9 per 1,000 persons in high-income countries and 3.6- 15.4 per 1,000 persons in low-income countries.<sup>1,3</sup> An epileptic seizure is caused by brain abnormal excessive and synchronous neuronal activity.<sup>4</sup> Epileptogenesis, a process of pathological changes that leads to epilepsy, is affected by altered gene expression (upregulation of mTOR), inflammation, protein production (serine/threonine protein kinase), and changes in

neuronal connectivity.<sup>5</sup>

Epilepsy could result in disability and lowered patients' quality of life.<sup>6</sup> The patient's quality of life with epilepsy is influenced by individual physical, mental, social, and demographic factors.<sup>7,8</sup> Compared to other chronic diseases and the general population, the quality of life of epileptic patients was found to be lower.<sup>8</sup> Patients with epilepsy often had low self-confidence and compromised social life.<sup>9</sup> Comorbidities also affect their quality of life.<sup>10</sup> Understanding the quality of life of a patient with epilepsy is important to assess epilepsy care from a patient's perspective and to evaluate treatment.<sup>11,12</sup> There were several questionnaires that were developed to determine the health-related quality of life of patients with epilepsy.<sup>13</sup>

Quality of Life in Epilepsy Inventory 10 (QOLIE-10) is a brief version of QOLIE-89. QOLIE-10 consists of three distinct topics, and these 10-item self-assessed questions are designed to be completed by patients.<sup>13,14</sup> Compared to

QOLIE-89 and QOLIE-31, which have 89 and 31 questions, respectively, QOLIE-10 is more compact and easier to complete. A valid (accurate) and reliable (repeatable) instrument is needed to portray the patient's quality of life accurately.<sup>15</sup> Indonesian version of QOLIE-10 is not available yet, making it challenging to be completed for patients with epilepsy in Indonesia. In this study, we translate QOLIE-10 to Indonesian and evaluate its validity and reliability.

## MATERIAL AND METHODS

### Study Design and Translation

This was a cross sectional study conducted at Neurology Clinic, Universitas Sebelas Maret, Surakarta, Central Java, Indonesia. Correspondence with QOLIE Development Group was done before starting this study. The original English version of QOLIE-10 was translated into Indonesian by a certified translator. The translated Indonesian version was then re-translated to English by another

## Survey Kualitas Hidup Pada Pasien Epilepsi:

## QOLIE-10-P

|                         |                   |
|-------------------------|-------------------|
| Tanggal ____/____/____. | Nama : _____.     |
|                         | Usia : ____ tahun |

## INSTRUKSI

QOLIE-10-P adalah sebuah survey singkat mengenai kualitas hidup sehubungan dengan kesehatan orang dewasa dengan epilepsi. Ada 10 pertanyaan mengenai kesehatan dan aktivitas sehari-hari, sebuah pertanyaan mengenai seberapa stress yang anda rasakan mengenai masalah-masalah dan kekhawatiran yang berhubungan dengan epilepsi, dan ulasan akan hal-hal yang paling mengganggu anda. Kuisioner ini seharusnya diisi/dilengkapi hanya oleh orang yang menderita epilepsi (bukan oleh saudara/teman) karena tidak ada orang lain yang tahu bagaimana anda merasakannya.

Jawab setiap pertanyaan dengan melingkari angka yang tepat (1,2,3,...). Jika anda tidak yakin bagaimana menjawab pertanyaan tersebut, tolong berikan jawaban terbaik anda dan tuliskan komentar atau penjelasan di samping halaman. Catatan-catatan ini mungkin bermanfaat jika anda mendiskusikan QOLIE-10-P dengan dokter anda. Melengkapi QOLIE-10-P sebelum dan sesudah perubahan pengobatan mungkin membantu anda dan dokter anda untuk memahami bagaimana

Pertanyaan-pertanyaan ini adalah tentang bagaimana selama ini anda MERASAKAN dan jenis-jenis masalah yang telah anda alami selama 4 minggu terakhir. Untuk masing-masing pertanyaan, tolong berikan satu jawaban yang paling mendekati dengan apa yang anda rasakan.

Seberapa sering selama 4 minggu terakhir.... (lingkari salah satu angka pada tiap baris)

|                                                                                                                                                  | Sepanjang waktu | Sebagian besar waktu        | Sering            | Kadang-kadang   | Jarang            | Tidak pernah                |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------------------|-------------------|-----------------|-------------------|-----------------------------|
| 1. Apakah anda punya banyak energi?                                                                                                              | 1               | 2                           | 3                 | 4               | 5                 | 6                           |
| 2. Pernahkah anda merasa sedih dan terpuruk?                                                                                                     | 1               | 2                           | 3                 | 4               | 5                 | 6                           |
| 3. Seberapa sering selama 4 minggu terakhir epilepsi atau obat antiepilepsi anda menyebabkan masalah dalam menyetir mobil (atau kendaraan lain)? |                 | Sangat terganggu            | Terganggu         | Cukup terganggu | Sedikit terganggu | Sama sekali tidak terganggu |
|                                                                                                                                                  |                 | 1                           | 2                 | 3               | 4                 | 5                           |
| Selama 4 minggu terakhir...                                                                                                                      |                 | Sama sekali tidak terganggu | Sedikit terganggu | Cukup terganggu | Terganggu         | Sangat terganggu            |
| 4. Seberapa besarlah keterbatasan anda dalam bekerja mengganggu anda?                                                                            |                 | 1                           | 2                 | 3               | 4                 | 5                           |
| 5. Seberapa besarlah keterbatasan-keterbatasan sosial anda mengganggu anda?                                                                      |                 | 1                           | 2                 | 3               | 4                 | 5                           |

| Selama 4 minggu terakhir...                                                    | Sama sekali tidak terganggu | Sedikit terganggu | Cukup terganggu | Terganggu           | Sangat terganggu        |
|--------------------------------------------------------------------------------|-----------------------------|-------------------|-----------------|---------------------|-------------------------|
| 6. Seberapa besar kesulitan-kesulitan ingatan anda mengganggu anda?            | 1                           | 2                 | 3               | 4                   | 5                       |
| 7. Seberapa besar dampak fisik dari obat antiepilepsi mengganggu anda?         | 1                           | 2                 | 3               | 4                   | 5                       |
| 8. Seberapa besar efek-efek psikologis dari obat antiepilepsi mengganggu anda? | 1                           | 2                 | 3               | 4                   | 5                       |
|                                                                                |                             | Sangat takut      | Cukup takut     | Tidak terlalu takut | Tidak takut sama sekali |
| 9. Seberapa takutnya anda untuk mengalami kejang selama 4 minggu ke depan?     | 1                           | 2                 | 3               | 4                   |                         |

10. Bagaimanakah KUALITAS HIDUP anda selama 4 minggu terakhir (yaitu, bagaimana hal-hal tersebut berlaku dalam kehidupan anda?)

(Lingkari satu nomor di tangga ini)

|                                         |   |
|-----------------------------------------|---|
| Sangat bagus : belum pernah seburuk ini | 1 |
| Lumayan bagus                           | 2 |
| Baik & buruk cukup seimbang             | 3 |
| Lumayan jelek                           | 4 |
| Sangat jelek : belum pernah separah ini | 5 |

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Teliti kembali semua pertanyaan yang telah anda jawab di atas, pikirkanlah dampak keseluruhan dari masalah-masalah tersebut terhadap kualitas hidup anda **dalam 4 minggu terakhir**. (Lingkari satu nomor).

11. Seberapa besar keadaan kualitas hidup yang berkaitan dengan epilepsi anda **membuat anda stress**?

1 2 3 4 5

Dengan mempertimbangkan **SELURUH** pertanyaan yang telah anda jawab, tolong **tunjukkan area-area** yang berhubungan dengan epilepsi anda yang paling **PENTING** bagi anda **SAAT INI**

12. Beri nomor topik-topik berikut dari '1' sampai '7' dengan '1' sama dengan topik yang paling penting dan '7' untuk topik yang paling tidak penting. Tolong gunakan tiap nomor hanya satu kali.
- ☐ A. Tenaga (kelelahan)
- ☐ B. Emosi (mood)
- ☐ C. Aktivitas sehari-hari (pekerjaan, berkendara, aktivitas sosial, dan aktivitas lain)
- ☐ D. Fungsi mental (berpikir, berkonsentrasi, ingatan)
- ☐ E. Efek-efek pengobatan (fisik, mental)
- ☐ F. Khawatir mengenai kejang (dampak kejang)
- ☐ G. Kualitas hidup keseluruhan

Salinan QOLIE-10-P ini disediakan oleh QOLIE Development Group. Kami berharap anda sukses dalam kehidupan anda dengan epilepsi.

Figure 1. Indonesian Version of QOLIE-10.

independent translator. To minimize bias, the backward translator to English was unaware of what the questionnaire was intended to measure. Several unclear wordings were found in the backward process and resolved by discussion of the expert committee. The layout and pictures on the English and Indonesian versions of QOLIE-10 are identical. The QOLIE-10 in the Indonesian version is shown in [Figure 1](#). The translated questionnaire was then tested on five epilepsy patients to verify that it was understandable by an Indonesian native speaker. Participants recruited were then asked to complete the Indonesian version of QOLIE-10. A month after patients completed the questionnaire, they were asked to retake it for the second time. We compared the first-time and second-time questionnaires from these five patients using an introductory dependent t-test and measured the correlation coefficient. We concluded that our questionnaire is understandable by Indonesian native speakers since the  $r$  value was 0.967 ( $p < 0.001$ ). This study was approved by the Medical Faculty of Universitas Sebelas Maret, Surakarta, Central Java, Indonesia, with Ethical Clearance No. 101/UN27.06.6.1/KEP/EC/2021.

### Participants

Participants of this study were outpatients of the Neurology Clinic, Universitas Sebelas Maret, Surakarta, Central Java, Indonesia. Purposive sampling was used in this study, where potential participants that met inclusion and exclusion criteria were included. Inclusion criteria for this study were: (1) aged  $>18$  years old; (2) could read, write, and not dependent on anyone to complete the questionnaire; (3) Indonesian native speaker; and (4) agreed to participate in this study. The exclusion criteria for this study were: (1) patient with severe hearing and visual disability; and (2) a patient with acute psychiatric, neurological, or medical symptoms that influence the quality of life.

### Data Analysis

Data obtained was analyzed using SPSS for Windows. To analyze QOLIE-10 validity, we evaluate the item validity of each question. Item validity was analyzed by

comparing each item score in the pretest and post-test with the total score from all participants using Pearson bivariate analysis to find the calculated  $r$ -value. Each item is valid if the calculated  $r$ -value is greater than the critical  $r$ -value. The critical  $r$ -value of 70 samples for a 5% significance level was 0.235. To assess its reliability, we calculate the internal consistency reliability coefficients with Cronbach's alpha test and test-retest reliability by comparing the pretest and post-test scores with the dependent t-test. Data is considered statistically significant if  $p < 0.05$ .

## RESULTS

This study was conducted at Neurology Clinic, Universitas Sebelas Maret, Surakarta, Central Java, Indonesia, from January to March 2022. A total of 70 epilepsy patients met the inclusion and exclusion criteria and were included in this study. The Indonesian version of the QOLIE-10 questionnaire was completed by participants and statistically analyzed to measure its validity and reliability. The characteristic of the participants is presented in [Table 1](#).

To analyze QOLIE-10 validity, we evaluate the item validity of each question. Item validity was analyzed by comparing each item with a total score with Pearson bivariate analysis to find the calculated  $r$ -value. Each item is valid if the calculated  $r$ -value is greater than the critical  $r$ -value, where the critical  $r$ -value for 70 samples was 0.235. Data is statistically significant if  $p < 0.05$ . The calculated  $r$ -value and  $p$ -value of each question from the pretest and post-test were presented in [Table 2](#), showing that each question is valid and statistically significant. The internal consistency reliability coefficients (Cronbach's alpha) were 0.938 and 0.909 for the pretest and post-test, respectively. We also analyzed the test-retest reliability by comparing the pretest and post-test scores with the dependent t-test. We observed a strong correlation with  $r = 0.871$  ( $p = 0.000$ )

## DISCUSSION

Quality of Life in Epilepsy Inventory (QOLIE) is a self-assessed questionnaire to evaluate epilepsy patients' quality

of life. Developed in 1995, the original version of QOLIE-89 was developed with a long list of 89 questions.<sup>16</sup> To make it easier to complete, an effort to simplify QOLIE-89 was started. A more compact QOLIE-31 and QOLIE-10, which, as the name suggested, consisting of 31 and 10 questions, were then developed.<sup>17,18</sup> QOLIE-89 was designed to assess 17 scales that determined the quality of life in epilepsy, whereas, on the other hand, QOLIE-10 assessed seven scales, i.e., seizure-worry, overall quality of life, emotional well-being, energy or fatigue, cognitive functioning, the physical effect of medication, mental effects of medication, driving, work, and social. A questionnaire with multi-item scales is thought to have higher reliability and validity, also more detailed compared to its shorter counterpart. However, measurements with shorter instruments allow healthcare providers to do assessments efficiently without compromising the quality of information collected. Patients are also more reluctant to complete a long questionnaire, potentially affecting the clinician-patient relationship.<sup>18,19</sup> Fortunately, a previous study by Cramer JA et al. found that the QOLIE-10 component correlated with the corresponding QOLIE-89 subscale studies.<sup>20</sup>

Quality of Life in Epilepsy Inventory 10 (QOLIE-10) has been translated into various languages and used widely. Study on the validity and reliability of QOLIE-10 versions of Thai, Korean, Bengali, Chinese and Turkish have been carried out, showing a valid and reliable questionnaire.<sup>13,19,21-23</sup> However, QOLIE-10 has never been translated into Indonesian. The guideline for developing a questionnaire by Tsang S et al. suggested that if a validated questionnaire that could assess our interest is available in a different language, translating rather than developing a new questionnaire is preferable.<sup>24</sup> In this study, the QOLIE-10 questionnaire was translated from English into Indonesian after permission was granted by the QOLIE Development Group. Translating consists of three distinct processes, i.e., establishing an expert committee, forward translation, and backward translation.<sup>24</sup> In this study, an expert committee consisting



of the neurologist and linguistic expert was formed. The translation process was done by certified translators at Universitas Sebelas Maret Language Center, Surakarta,

Central Java, Indonesia.

Tsang S et al. suggested that forward translation to the target language should be done by at least two independent translators, preferably translators that understand the context and concept of the questionnaire.<sup>24</sup> Rather than clinicians, our translators were linguistic experts. To meet a common understanding of the questionnaire, we had an intensive discussion on epilepsy and the QOLIE-10 questionnaire before the translation process. After being translated to Indonesian, it was then re-translated to English by different translators to improve the accuracy of the translation process. This process was done by two translators, different from the translators done the forward translation. However, different from what Tsang S et al. suggested, we did not use a translator who had English as their mother language.<sup>24,25</sup> To minimize bias, the backward translator to English was unaware of what the questionnaire was intended to measure. Several unclear wordings were found in the backward process and resolved by discussion of the expert committee. After testing on five epilepsy patients, we concluded that our questionnaire was understandable by an Indonesian native speaker. Then, 70 patients were asked to complete the questionnaire twice, and the post-test was done approximately one month after the pretest. Considering when to do a post-test is crucial. Patients might still remember their first response if the interval is too short, or their condition might change if the interval is too long.<sup>24,26</sup>

Validity refers to what extent the results measure what the questionnaire is supposed to measure. In this study, we use Pearson bivariate analysis to assess the item validity of a questionnaire. The r-value obtained from the Pearson bivariate test was then compared with the critical r-value with a 5% significance level. Item is considered valid if the r-value is greater than the critical r-value with a 5% significance level.<sup>24,27</sup> On Pearson bivariate analysis, we observed an r-value ranging from 0.247 to 0.669 on the pretest and 0.396 to 0.668 on the post-test. All r-value is greater than 0.235, the critical r-value for 70 samples for 5% significance level, and all question had a p-value of <0.05.

**Table 1. Subject characteristic.**

| Subject Characteristics             | Total (n=70) |
|-------------------------------------|--------------|
| Age (Years), n (%)                  |              |
| 18-32                               | 26 (37.2)    |
| 33-54                               | 33 (47.1)    |
| 55-72                               | 11 (15.7)    |
| Gender, n (%)                       |              |
| Male                                | 28 (40)      |
| Female                              | 42 (60)      |
| Marital Status, n (%)               |              |
| Single                              | 21 (30.0)    |
| Married                             | 46 (65.7)    |
| Divorced or widowed                 | 3 (4.3)      |
| Educational Level, n (%)            |              |
| No formal education                 | 1 (1.4)      |
| Primary school                      | 4 (5.7)      |
| Secondary school                    | 8 (11.5)     |
| High school                         | 37 (52.8)    |
| Academy                             | 8 (11.5)     |
| University                          | 12 (17.1)    |
| Occupation, n (%)                   |              |
| Student                             | 10 (14.2)    |
| Employee                            | 12 (17.1)    |
| Housewife                           | 22 (31.5)    |
| Not working                         | 18 (25.7)    |
| Entrepreneur                        | 5 (7.2)      |
| Others                              | 3 (4.3)      |
| Monthly Income, n (%)               |              |
| No income                           | 5 (7.2)      |
| <Rp. 1.000.000                      | 31 (44.3)    |
| Rp. 1.000.000 - Rp. 5.000.000       | 24 (34.3)    |
| >Rp. 5.000.000                      | 10 (14.2)    |
| Seizure Frequency, n (%)            |              |
| Seizure free $\geq 2$ years         | 6 (8.6)      |
| Seizure free 1-2 years              | 9 (12.8)     |
| Occasional (<1 seizure/year)        | 18 (25.7)    |
| Moderate (1-11 seizures/year)       | 26 (37.2)    |
| Frequent ( $\geq 1$ seizure/month)  | 11 (15.7)    |
| Seizure Type, n (%)                 |              |
| Focal onset                         | 27 (38.5)    |
| General onset                       | 38 (54.3)    |
| Unknown                             | 5 (7.2)      |
| Last Seizure, n (%)                 |              |
| <1 month                            | 7 (10.0)     |
| 1 to <12 months                     | 37 (52.8)    |
| 12 to <24 months                    | 18 (25.7)    |
| $\geq 24$ months                    | 8 (11.5)     |
| Epilepsy Type, n (%)                |              |
| Idiopathic                          | 49 (70.0)    |
| Symptomatic                         | 21 (30.0)    |
| Anti-Epileptic Drug Quantity, n (%) |              |
| Monotherapy                         | 46 (65.7)    |
| Polytherapy                         | 24 (34.3)    |
| Anti-Epileptic Drug Type, n (%)     |              |
| Phenytoin                           | 10 (14.2)    |
| Valproic acid                       | 25 (35.8)    |
| Carbamazepine                       | 11 (15.7)    |
| Combination                         | 24 (34.3)    |

**Table 2.** The r-value and p-value of each question from the pretest and post-test.

| Item                              | Pretest |        | Post-test |        |
|-----------------------------------|---------|--------|-----------|--------|
|                                   | r       | p      | r         | p      |
| Energy/fatigue                    | 0.416   | 0.000* | 0.446     | 0.001* |
| Emotional well-being              | 0.247   | 0.039* | 0.438     | 0.001* |
| Driving                           | 0.272   | 0.023* | 0.456     | 0.000* |
| Work                              | 0.588   | 0.000* | 0.553     | 0.000* |
| Social                            | 0.620   | 0.000* | 0.668     | 0.000* |
| Cognitive functioning             | 0.613   | 0.000* | 0.621     | 0.000* |
| The physical effect of medication | 0.669   | 0.000* | 0.648     | 0.000* |
| The mental effect of medication   | 0.464   | 0.000* | 0.613     | 0.000* |
| Seizure worry                     | 0.397   | 0.001* | 0.466     | 0.000* |
| The overall quality of life       | 0.272   | 0.023* | 0.396     | 0.033* |

\*Statistically significant if p-value less than 0.05

This finding indicates that the Indonesian version of QOLIE-10 is valid.

Reliability is defined as to what extent the results can be reproduced when the research is repeated under the same conditions. Reliability could be assessed by checking the consistency of results across time (test-retest reliability), across different observers (interrater reliability), and across parts of the test itself (internal consistency).<sup>24,28,29</sup> Two components of reliability, internal consistency, and test-retest reliability, were applicable in this study and tested to the Indonesian version of the QOLIE-10 questionnaire. Cronbach alpha test was used to assess internal consistency, and we found coefficients of 0.938 and 0.909 for the pretest and post-test, respectively, with a p-value of <0.05. Cronbach alpha value of >0.7 is considered reliable.<sup>28,29</sup> Dependent t-test was done to determine test-retest reliability by comparing pretest and post-test scores, and we observed a correlation coefficient with  $r=0.871$  ( $p=0.000$ ). A correlation coefficient >0.8 is considered very strong.<sup>28,29</sup> From internal consistency and test-retest reliability result, we can conclude that the Indonesian version of QOLIE-10 is reliable. Barranco-Camargo LA et al., who analyzed the feasibility, reliability, and validity of the QOLIE-10 in adult refractory epilepsy patients in the Colombian Caribbean, also found that this questionnaire is valid and reliable.<sup>30</sup> Similarly, Mollaoğlu M et al. have reported that the Turkish version of QOLIE-10 had excellent item and test-retest reliability.<sup>31</sup>

This is the first study to translate the QOLIE-10 questionnaire into Indonesian

and evaluate its validity and reliability. The Indonesian version of QOLIE-10 we develop is understandable to Indonesian native speakers, valid, and reliable. This result indicates that it could be used to assess epilepsy patients' quality of life in Indonesia. However, this study had a relatively small number of subjects and backward translation not done by a translator whose mother language is English. Besides, test-retest reliability has not been assessed for this questionnaire due to limited time and resources. This assessment is needed to determine the stability of the questionnaire by conducting an assessment at least twice on the same subject. Further study comparing the Indonesian version of QOLIE-10 with QOLIE-31 and QOLIE-89 is also needed to confirm that the shorter version of the questionnaire had a comparable quality of information to its longer version.

## CONCLUSION

QOLIE-10 is understandable to Indonesian native speakers, valid, and reliable, making it a practical questionnaire to use in the clinical setting.

## CONFLICT OF INTEREST

The authors declare that the study was conducted without commercial or financial relationships that could be considered a potential conflict of interest.

## ETHICAL CONSIDERATION

This study was approved by the Medical Faculty of Universitas Sebelas Maret, Surakarta, Central Java, Indonesia, with

Ethical Clearance No. 101/UN27.06.6.1/KEP/EC/2021.

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## AUTHOR CONTRIBUTION

All authors contributed to the study from the conceptual framework, data gathering, and analysis until the study's results were interpreted upon publication.

## REFERENCES

- Beghi E. The Epidemiology of Epilepsy. *Neuroepidemiology*. 2020;54(2):185–91.
- Fisher RS, Acevedo C, Arzimanoglou A. ILAE Official Report: A practical clinical definition of epilepsy. *Epilepsia*. 2014;55(4):475–482.
- Beghi E, Giussani G. Aging and the Epidemiology of Epilepsy. *Neuroepidemiology*. 2018;51(3–4):216–223.
- Shellhaas RA. Seizure classification, etiology, and management. *Handb Clin Neurol*. 2019;162:347–361.
- Manford M. Recent advances in epilepsy. *J Neurol*. 2017;264(8):1811.
- Beghi E, Giussani G, Abd-Allah F, Nichols E, Abdela J, Abdelalim A, et al. Global, regional, and national burden of epilepsy, 1990–2016: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet Neurol*. 2019;18(4):357–375.
- Baranowski CJ. The quality of life of older adults with epilepsy: A systematic review. *Seizure - Eur J Epilepsy*. 2018;60:190–197.
- Viteva EI. Seizure frequency and severity: How really important are they for the quality of life of patients with refractory epilepsy. *Ann Indian Acad Neurol*. 2014;17(1):35–42.
- Ashjzadeh N, Yadollahikholes G, Ayoobzadehshirazi A, Sadraei N, Hadi N. Comparison of the health-related quality of life between epileptic patients with partial and generalized seizure. *Iran J Neurol*. 2014;13(2):94–100.
- Vu LC, Piccenna L, Kwan P, O'Brien TJ. New-onset epilepsy in the elderly. *Br J Clin Pharmacol*. 2018;84(10):2208.
- Singh P, Pandey AK. Quality of life in epilepsy. *Int J Res Med Sci*. 2016;5(2):452–455.
- Pimpalkhute SA, Bajait CS, Dakhale GN, Sontakke SD, Jaiswal KM, Kinge P. Assessment of quality of life in epilepsy patients receiving anti-epileptic drugs in a tertiary care teaching hospital. *Indian J Pharmacol*. 2015;47(5):551–554.
- Mollaoğlu M, Mollaoğlu M, Durna Z. Validity and Reliability of the Quality of Life in Epilepsy Inventory (QOLIE-10) For Turkey. *Arch Neuropsychiatry*. 2017;54(3):239–243.
- Cramer JA, Perrine K, Devinsky O, Meador K. A Brief Questionnaire to Screen for Quality

- of Life in Epilepsy The QOLIE-10. *Epilepsia*. 1996;37(6):577–582.
15. Blake MR, Raker JM, Whelan K. Validity and reliability of the Bristol Stool Form Scale in healthy adults and patients with diarrhoea-predominant irritable bowel syndrome. *Aliment Pharmacol Ther*. 2016;44(7):693–703.
  16. Devinsky O, Vickrey BG, Cramer J, Perrine K, Hermann B, Meador K, et al. Development of the Quality of Life in Epilepsy Inventory. *Epilepsia*. 1995;36(11):1089–1090.
  17. Cramer JA, Perrine K, Devinsky O, Bryant-Comstock L, Meador K, Hermann B. Development and cross-cultural translations of a 31-item quality of life in epilepsy inventory. *Epilepsia*. 1998;39(1):81–88.
  18. Cramer JA, Arrigo C, Van Hammée G, Bromfield EB. Comparison between the QOLIE-31 and derived QOLIE-10 in a clinical trial of levetiracetam. *Epilepsy Res*. 2000;41(1):29–38.
  19. Lee SA, Yun SC, Lee BI. Reliability and Validity of the Korean Version of QOLIE-10 in Epilepsy. *J Clin Neurol*. 2006;2(4):238–345.
  20. Cramer JA, Perrine K, Devinsky O. A Brief Questionnaire to Screen for Quality of Life in Epilepsy The QOLIE-10. 1996;37(6):577–582.
  21. Kanjanasilp J, Khaewichit S, Richards RME, Preechagoon Y. Thai Version of the Quality-of-Life in Epilepsy Inventory: Comparison Between the QOLIE-31 and the QOLIE-10. *Journal*. 2004;3(1):35–42.
  22. Basu S, Sanyal D, Ghosal M, Roy B, Senapati AK, Das SK. Psychometric properties of Bengali version of QOLIE-10 in epileptic patients. *Ann Indian Acad Neurol*. 2008;11(1):28–32.
  23. Gao L, Xia L, Pan SQ, Xiong T, Li SC. Reliability and validity of QOLIE-10 in measuring health-related quality of life (HRQoL) in Chinese epilepsy patients. *Epilepsy Res*. 2014;108(3):565–575.
  24. Tsang S, Royse CF, Terkawi AS. Guidelines for developing, translating, and validating a questionnaire in perioperative and pain medicine. *Saudi J Anaesth*. 2017;11(5):S80–S89.
  25. Lestari ND, Sary NL, Ummah AK, Zulkarnain Z, Astini N. Description of depression in people with epilepsy in Aceh. *Bali Medical Journal*. 2021;10(2):521–525.
  26. Golden N, Mardhika PE, Nirvana W, Sukarya IM, Prabawa IPY. Risk factors and novel prognostic score for predicting the 14-day mortality of severe traumatic brain injury patients. *Intisari Sains Medis*. 2020;11(3):1020–1028.
  27. Gunadharma S, Nurimaba N, Rochayatin O. Uji Kesahihan Dan Keandalan Quality of Life in Epilepsy Inventory (Qolie-31) Versi Bahasa Indonesia. *Neurona*. 2015;32(3):1–9.
  28. Bolarinwa OA. Principles and methods of validity and reliability testing of questionnaires used in social and health science research. *Niger Postgrad Med J*. 2015;22(4):195–201.
  29. Taherdoost H. Validity and Reliability of the Research Instrument; How to Test the Validation of a Questionnaire/Survey in a Research. *SSRN Electron J*. 2016;5(3):28–36.
  30. Barranco-Camargo LA, Usta-Agamez E, López-García J, Juado-Lopez S, Zabala-Caraballo C, Ramos-Clason E. Validity and reliability of the QOLIE-10 instrument for assessing health related quality of life in epilepsy of refractory epilepsy adult patients at a Colombian neurological center. *Rev Neurol*. 2019;61(12):473–480.
  31. Mollaoğlu M, Mollaoğlu M, Durna Z. Validity and reliability of the quality of life in epilepsy inventory (QOLIE-10) for Turkey. *Noropsikiyatri Ars*. 2017;54(3):239–243.



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